

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B5: GENERAL RELATIVITY

TRINITY TERM 2024

Monday, 10 June, 2:30 pm – 4:30 pm

*Answer **two** questions.*

*Start the answer to each question in a **fresh book**.*

The use of approved calculators is permitted.

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

Page 2 contains physics formulae and data for this paper.

The questions start on page 3.

Expressed in terms of the Christoffel symbols $\Gamma_{\beta\gamma}^\alpha$, the Riemann tensor has components

$$R^\gamma_{\alpha\beta\delta} = \frac{\partial}{\partial x^\delta} \Gamma_{\alpha\beta}^\gamma - \frac{\partial}{\partial x^\beta} \Gamma_{\alpha\delta}^\gamma + \Gamma_{\alpha\beta}^\epsilon \Gamma_{\delta\epsilon}^\gamma - \Gamma_{\alpha\delta}^\epsilon \Gamma_{\beta\epsilon}^\gamma.$$

The Ricci tensor is defined as $R_{\alpha\beta} \equiv -R^\gamma_{\alpha\beta\gamma}$. When expressed in a basis in which the metric tensor $g_{\alpha\beta}$ is diagonal, it can be written as

$$R_{\alpha\beta} = \frac{1}{2} \frac{\partial^2}{\partial x^\alpha \partial x^\beta} \ln |g| - \frac{\partial \Gamma_{\alpha\beta}^\gamma}{\partial x^\gamma} + \Gamma_{\alpha\gamma}^\delta \Gamma_{\beta\delta}^\gamma - \frac{1}{2} \Gamma_{\alpha\beta}^\gamma \frac{\partial}{\partial x^\gamma} \ln |g|,$$

where $|g|$ is the modulus of the determinant of $g_{\alpha\beta}$.

The Einstein equation is

$$R_{\alpha\beta} - \frac{1}{2} g_{\alpha\beta} R - \Lambda g_{\alpha\beta} = -\frac{8\pi G}{c^4} T_{\alpha\beta},$$

where $R \equiv g^{\alpha\beta} R_{\alpha\beta}$ is the Ricci scalar, Λ is the cosmological constant and $T_{\alpha\beta}$ is the energy-momentum tensor.

For the line element

$$ds^2 = -c^2 dt^2 + R^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2 \right]$$

the Einstein equation can be written as the pair of equations

$$\begin{aligned} \ddot{R} &= -\frac{4\pi G}{3} \left(\rho + \frac{3p}{c^2} \right) R + \frac{1}{3} \Lambda c^2 R, \\ \dot{R}^2 &= \frac{8\pi G}{3} \rho R^2 + \frac{1}{3} \Lambda c^2 R^2 - c^2 k, \end{aligned}$$

where ρ and p are the rest-frame density and pressure, respectively.

1. The energy-momentum tensor for a perfect fluid is

$$T^{\alpha\beta} = P g^{\alpha\beta} + \left(\rho + \frac{P}{c^2} \right) U^\alpha U^\beta,$$

where ρ is the mass density, P is the pressure and U^α is the fluid velocity such that $U_\alpha U^\alpha = -c^2$. Assume the $T^{\alpha\beta}$ is covariantly conserved,

$$T^\alpha_{\beta;\alpha} = 0.$$

a) Find a current, j^α , which satisfies the conservation equation

$$j^\alpha_{;\alpha} = U^\alpha P_{,\alpha}.$$

[6]

b) Show that the motion of a pressureless fluid in a gravitational field that satisfies conservation of energy-momentum obeys the geodesic equation,

$$U^\alpha U^\beta_{;\alpha} = 0.$$

[4]

Consider the following set of equations

$$R_{\alpha\beta} - \frac{1}{n} R g_{\alpha\beta} = 0.$$

Explain why, for $n = 2$, you expect 10 equations. Take the trace of these equations to find the value of n for which there are only 9 (and not 10) equations. Use the contracted Bianchi identities,

$$R_{,\alpha} - 2R^\beta_{\alpha;\beta} = 0,$$

to show that R has to be a constant.

[5]

d) Consider a modified version of the Einstein field equations,

$$R_{\alpha\beta} - \frac{1}{4} R g_{\alpha\beta} = -\frac{8\pi G}{c^4} \left(T_{\alpha\beta} - \frac{1}{4} T g_{\alpha\beta} \right),$$

where $T = T^\alpha_\alpha$. Use the contracted Bianchi identities, and conservation of energy and momentum to show that this equation implies the normal Einstein field equations with an additional cosmological constant term.

[6]

e) In a universe where $T_{\alpha\beta} = 0$, find the value of the cosmological constant for a space-time whose Riemann tensor is given by

$$R_{\alpha\beta\mu\nu} = \frac{K}{12} (g_{\alpha\mu} g_{\beta\nu} - g_{\alpha\nu} g_{\beta\mu}).$$

[4]

2. The metric of a charged black hole is given by

$$ds^2 = -A(r)c^2 dt^2 + \frac{1}{A} dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2),$$

where

$$A(r) = 1 - \frac{2GM}{c^2 r} + \frac{GQ^2}{c^4 r^2},$$

M is the mass, and Q is the charge of the black hole.

a) Show that for a massive particle in an equatorial orbit, the following equations are satisfied:

$$At = k, \quad r^2 \dot{\phi} = h,$$

where k and h are constants, $\dot{X} = dX/d\tau$, and τ is proper time. [4]

b) Define the Schwarzschild radius $r_G = 2GM/c^2$ and the charge radius, $r_Q = \sqrt{G}Q/c^2$. Show that, on an equatorial orbit, the particle obeys

$$\frac{1}{2} \dot{r}^2 + V_{\text{eff}}(r) = \frac{1}{2} c^2 (k^2 - 1),$$

where

$$V_{\text{eff}}(r) = -\frac{c^2 r_G}{2r} + \frac{h^2 + c^2 r_Q^2}{2r^2} - \frac{h^2 r_G}{2r^3} + \frac{(hr_Q)^2}{2r^4}.$$

[5]

c) Consider now a massive particle that starts from rest at infinity. Show that it comes to a stop at a certain distance, r_* from the centre of the black hole and determine the value of r_* . What happens after it stops at r_* ? [6]

d) Consider the case where $r_Q > r_G$. A freely falling particle starts from rest at $r = r_+$. Show that

$$\dot{r}^2 = c^2 \frac{r_G(r_+ - r_*)}{r_+^2 r^2} (r_+ - r)(r - r_-),$$

where $r_- = r_+ r_*/(r_+ - r_*)$. Describe the motion of such a particle. How does that differ from what you would find with a Schwarzschild black hole? [6]

e) Show that the proper time, τ , elapsed between the particle traversing from r_+ to r_- is

$$\tau = \left(\frac{r_+ + r_-}{r_G} \right)^{3/2} \frac{\pi r_G}{2}.$$

[Hint: Recall that $\int_{-1}^1 \frac{dx}{\sqrt{1-x^2}} = \pi$.]

[4]

3. The metric of an anisotropic universe is given by

$$ds^2 = -c^2 dt^2 + e^{2\alpha} dx^2 + e^{2\beta} (dy^2 + dz^2),$$

where α and β are functions of t only. The energy-momentum tensor is of the form

$$T^\mu_\nu = \begin{pmatrix} -\rho c^2 & 0 & 0 & 0 \\ 0 & w_A \rho c^2 & 0 & 0 \\ 0 & 0 & w_B \rho c^2 & 0 \\ 0 & 0 & 0 & w_B \rho c^2 \end{pmatrix},$$

where ρ is the mass density, w_A and w_B are constant equations of state, where μ and ν run from 0 to 3 with $x^0 = ct$.

a) Show that the Christoffel symbols for this space time are:

$$\Gamma^0_{xx} = \frac{\dot{\alpha}}{c} e^{2\alpha}, \quad \Gamma^0_{yy} = \Gamma^0_{zz} = \frac{\dot{\beta}}{c} e^{2\beta}, \quad \Gamma^x_{0x} = \frac{\dot{\alpha}}{c}, \quad \Gamma^y_{0y} = \Gamma^z_{0z} = \frac{\dot{\beta}}{c}.$$

[5]

b) Use conservation of energy-momentum to find an evolution equation for ρ and show that

$$\rho(t) \propto e^{-(1+w_A)\alpha(t)-2(1+w_B)\beta(t)}.$$

[3]

c) Use the definition of the Ricci tensor to show that the Einstein Field equations can be written as:

$$\begin{aligned} \dot{\beta}^2 + 2\dot{\alpha}\dot{\beta} &= 8\pi G\rho \\ 2\ddot{\beta} + 3\dot{\beta}^2 &= -8\pi Gw_\alpha\rho \\ \ddot{\alpha} + \ddot{\beta} + \dot{\alpha}^2 + \dot{\beta}^2 + \dot{\alpha}\dot{\beta} &= -8\pi Gw_\beta\rho. \end{aligned}$$

[6]

d) Show that, for $w_A = w_B = -1$, ρ is constant. Use the field equations to show that the expansion is isotropic (i.e. that $\dot{\alpha} = \dot{\beta}$) and exponential. By looking at the fixed point (where $\ddot{\alpha} = \ddot{\beta} = 0$) and linearising the field equations, show that isotropic expansion is stable.

[4]

e) Use d) to show that, in the stable, isotropic, limit, the physical volume of a fixed comoving volume of space evolves as

$$V \propto e^{3qt}.$$

[3]

f) Consider now the case where $w_A \neq w_B$ and the expansion is anisotropic. Devise two ways in which you might use observations to detect and constrain the anisotropy of the Universe.

[4]

4. Consider two masses of equal value, μ , lying on the z – axis with a relative displacement between them given by

$$\Delta z(t) = L + a \cos(\omega t).$$

a) What physical configuration might this correspond to? Calculate the quadrupole moment of this configuration

$$I^{ij} = \int d^3r T^{00}(t, \vec{r}) r^i r^j,$$

where $T^{\mu\nu}$ is the energy momentum tensor of the system.

[5]

b) Assume we are in flat space and can expand the metric as

$$g_{\alpha\beta} = \eta_{\alpha\beta} + h_{\alpha\beta},$$

where $\eta_{\alpha\beta}$ is the Minkowski metric and the components of $h_{\alpha\beta}$ are small. Define

$$\bar{h}_{\alpha\beta} = h_{\alpha\beta} - \frac{h}{2} \eta_{\alpha\beta},$$

where $h = h^\alpha{}_\alpha$. Find an expression for the transverse-traceless $\bar{h}_{ij}^{TT}$ propagating along the x-axis, in terms of m and ω .

[Hint: recall that the transverse part of a spatial three tensor, A_{ij} , is $A_{ij}^{TT} = (P_{ia}P_{jb} - \frac{1}{2}P_{ij}P_{ab})A_{ab}$ where $P_{ij} = \delta_{ij} - n_i n_j$ and $\vec{n}$ is the unit vector along the direction of propagation.]

[7]

c) Consider a small ring, of radius δR , of test masses far away from the source, centred on, and perpendicular to the x-axis. Assume that the source has been oscillating for a long time. Describe how the test masses will move.

[4]

d) The radial energy flux of gravitational waves is given by

$$\mathcal{F}_r = -\frac{c^4}{32\pi G} \left\langle \frac{\partial}{\partial t} h_{ij}^{TT} \frac{\partial}{\partial r} h_{ij}^{TT} \right\rangle,$$

where $\langle \dots \rangle$ corresponds to averaging over one period, $T = 2\pi/\omega$. Find the total gravitational wave luminosity for this system.

[5]

e) Find the total energy radiated over one period. Write your expressions in terms of the maximum velocities and discuss the dependence on L/a .

[4]